

Infosys Internship — Individual Contribution Documentation

Milestone 2: Week 3 & 4 — Risk Prediction & Readmission Forecasting

Contributor: Penchala Prasad | Project: St. Jude Medical Center — HealthForecast AI | Milestone: 2

1. Executive Summary & Overview

During Milestone 2 (Week 3 & 4), the core objective focused on engineering predictive patient risk intelligence systems and readmission forecasting workflows. St. Jude Medical Center's HealthForecast AI platform was upgraded with client-side ML classifiers, real-time risk simulators, temporal horizon forecasting, and downloadable decision support reports.

2. Key Modules & Individual Contribution

As an individual contributor, I designed, built, and verified the following clinical modules:

- **Machine Learning Telemetry Integration:** Configured Gradient Boosting model telemetry presenting 97.5% classification accuracy, 0.9867 F1 score, and 0.9960 ROC-AUC metric visualizers.
- **Real-Time Patient Risk Score Simulator:** Implemented an interactive clinical calculator allowing clinicians to adjust Length of Stay, Emergency Visits, Comorbidities, and Medication status to view instant readmission risk scores.
- **Readmission Risk Dashboards:** Built cohort risk distribution donut charts, stacked diagnosis risk bars, and priority patient triage queues.
- **Temporal Readmission Forecasting Workflows:** Developed 30-Day, 60-Day, and 90-Day horizon projection area charts alongside departmental trajectory tables.
- **Forecasting CSV Report Exporter:** Created client-side report export utilities for clinical decision makers.
- **Clinical Decision Support (CDS) Protocols:** Built care checklist modules, diet recommendations, and post-discharge guidelines.

■ Milestone 2 Outcomes Verification Matrix

Required Task / Deliverable	Target Outcome	Implementation Component	Status
Train patient risk prediction models	Model accuracy (97.5%) & feature importance	/doctor/risk-predictions	COMPLETED
Generate patient risk scores	Interactive real-time risk score simulator	/doctor/risk-predictions	COMPLETED
Build risk prediction dashboards	Risk stratification donut & triage table	/doctor/risk-predictions	COMPLETED
Develop readmission forecasting	30/60/90-day horizon projection charts	/doctor/readmission	COMPLETED
Generate forecasting reports	Automated CSV report exporter	/doctor/readmission	COMPLETED
Build clinical insights modules	CDS care protocols & discharge checklist	/doctor/clinical-insights	COMPLETED

